Al-Balqa' Applied University
Faculty of Engineering Technology
Electronics1
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Mid Exam
2022/2023

جامعـة البلقاء التطبيقيـة
كلية الهندسة التكنولوجية
Time: 50 min.
/11/2022

اسم الطالب : أ مثنى خالد الطويل
التخصص : هندسة حاسوب

1- Each atom in a silicon crystal has:
   a four valence electrons
   b- four conduction electrons
   c- eight valence electrons, four of its own and four shared
   d- no valence electrons because all are shared with other atoms

2- When a voltmeter is placed across a forward-biased diode, it will read a voltage approximately equal to:
   a- the bias battery voltage   b- 0 V
   c- the diode barrier potential   d- the total circuit voltage

*** For forward bias for the silicon diode given $I_D = 4$ mA at 0.58 V answer the question (3and 4)**

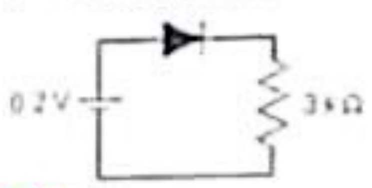
3- the ac resistance at given current equal
   a- 6,5 Ω   b- 13 Ω   c- 145 Ω   d- 290 Ω

4-The dc resistance at this current level is
   a- 6.5 Ω   b- 13 Ω   c- 145 Ω   d- 290 Ω

5- The depletion region consists of
   a- nothing but minority carrier's   b- positive and negative ions
   c- No majority carrier's   d- answer (b) and (c)
   e- Neither (a).(b),nor (c)

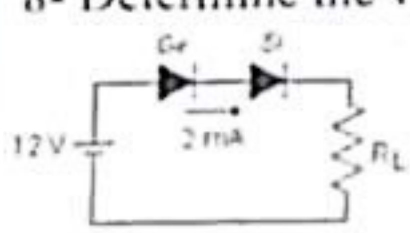
6- If you are checking a 50 Hz full-wave bridge rectifier and observe that the output has a 50 Hz ripple
   a- the circuit is working properly   b- there is an open diode
   c- the transformer secondary is shorted   d- the filter capacitor is leakage

7- Determine the voltage across the resistor 3 kΩ.



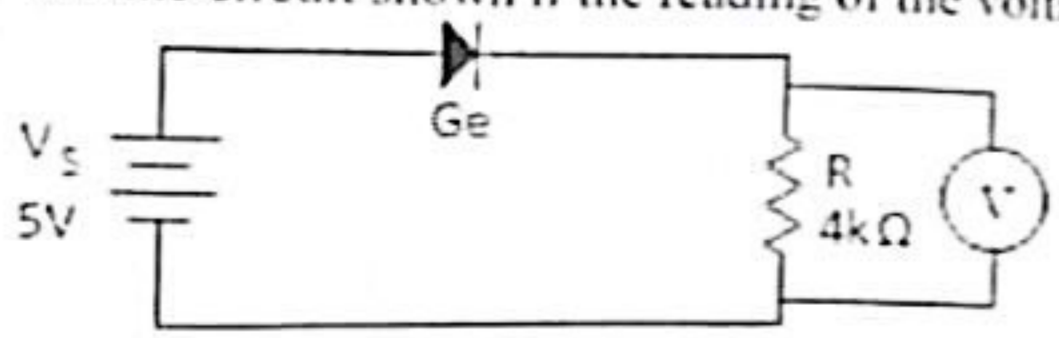
a- 0 V   b- 0.9V   c-0.2V   d- 0.44V

8- Determine the value of the load resistor.



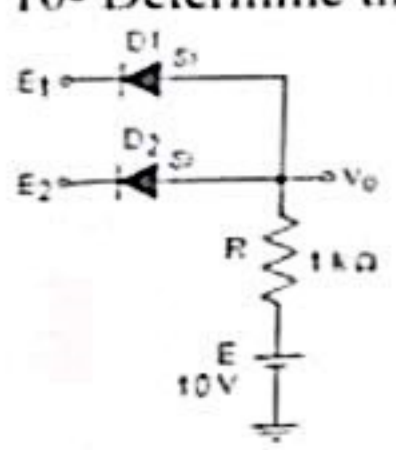
a- 5 kΩ   b- 6 kΩ   c- 5.5 kΩ   d-11 kΩ

9- for the circuit shown if the reading of the voltmeter is 5 V so the diode is:



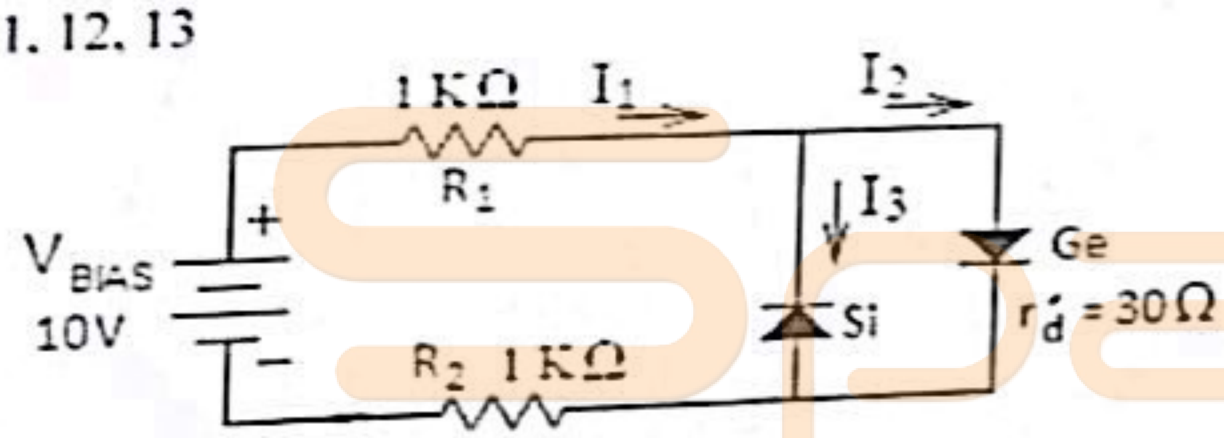
a- in normal operation b- open c- short d- cannot be determined

10- Determine the current through each diode if E1 = E2 = 0 V.



a-0.7 mA b-4.65 mA c-9.3 mA d-18.6 mA

*- for the circuit shown, assume that both diodes are complete piecewise (including r_d). Answer questions 11, 12, 13

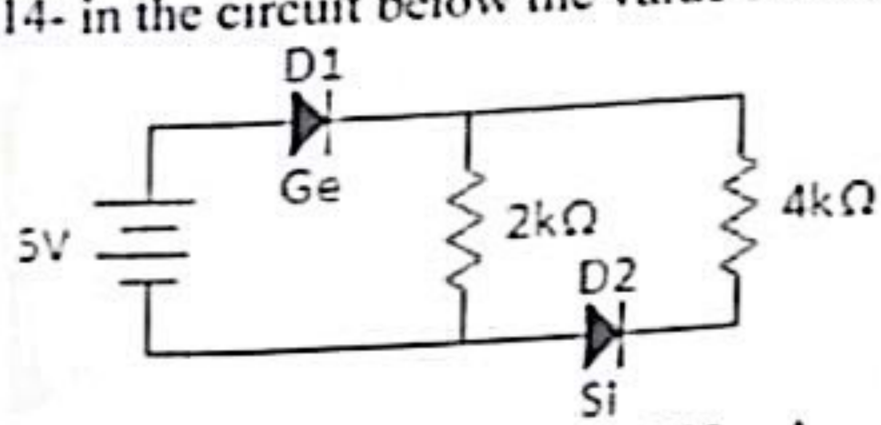


11- the value of the current I_1 equal
a-0 mA b-4.78 mA c-5.58 mA d-4.65 mA e- 5 mA

12- the value of I_2 equal
a-0 mA b-4.78 mA c-5.58 mA d-4.65 mA e- 5 mA

13- the value of I_3 equal
a-0 mA b-4.78 mA c-5.58 mA d-4.65 mA e- 5 mA

14- in the circuit below the value of the current through the resistor 4 kΩ equal



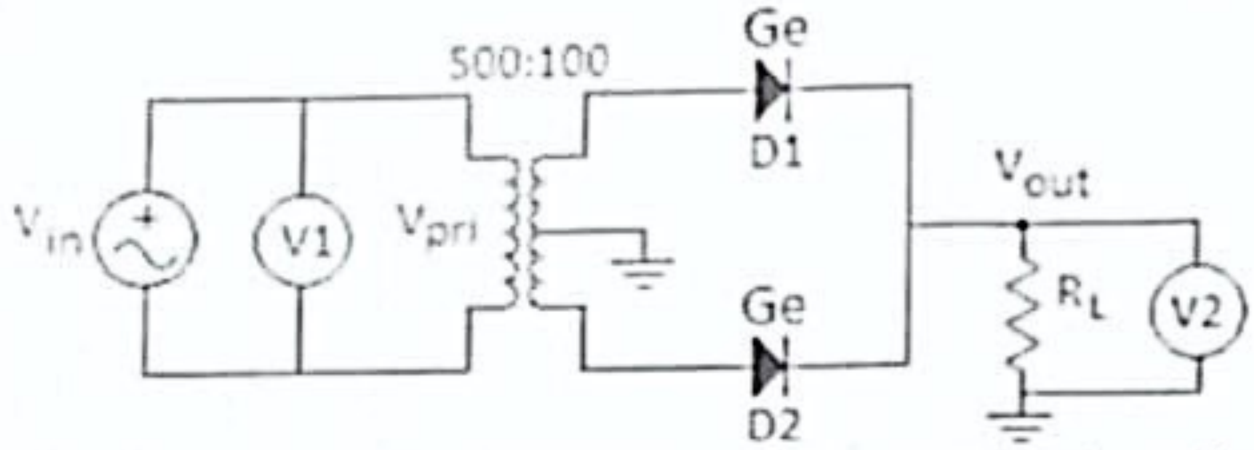
a-0 mA b-1.075 mA c-1 mA d-1.25 mA

15- the voltage across D2 in circuit question 14 is

a-0 V b- 5V c-4.3 V d-0.7 V

Correct Answer : 4.7

* **For the circuit shown answer the questions 16, 17, 18, 19, 20, 21**



16- If the prak value of the output is (V_{pout} =2.2 V) the rading of the voltmeter V2 is

a-2.74 V b- 1.4 V c-0.68 V d-3.04 V

17- The maximum value of the secondary voltage (V_{sec}) equal to

a-8.6 V **b- 15 V** c-5 V d-10 V

18- The maximum value of the primary voltage (V_{pri}) equal to

a-5 V b- 35 V c-45 V d-25 V

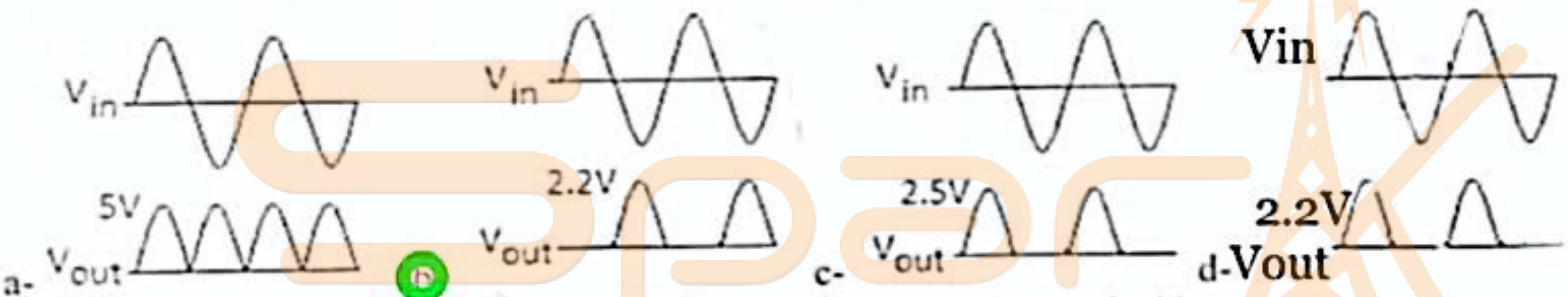
19-The reading of the voltmeter V1 is

a-35.35 V b- 17.68 V c-42.43 V d-21.21 V

20-PIV for each diode is

~~a-10 V b-6.2 V c-9.3 V d-4.7 V~~ Correct Answer: 9.7

21- If in the circuit the diode D1 is open so the output pulses will be



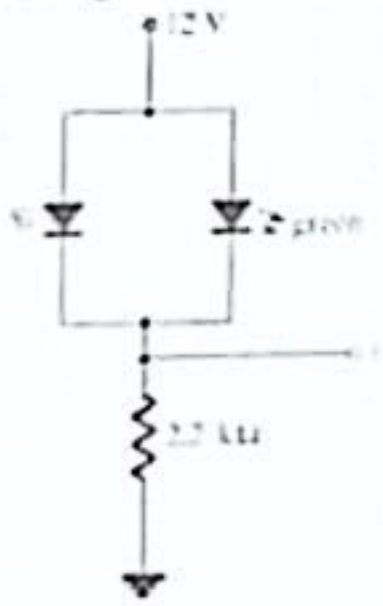
22- for practical zener diode in forward bias the voltge across its treminal is

a- 0 V b- the applied voltage c- 0.7 V d- the Zener diode voltage

23- By addition of pentavalent to the silicon material the hole current is became

a- Majority carrier b- Minority Carrier c- No change the hole current

24- For the circuit shown, if the forward voltage of the green diode is (Vd =2V) the value of the current through the LED is



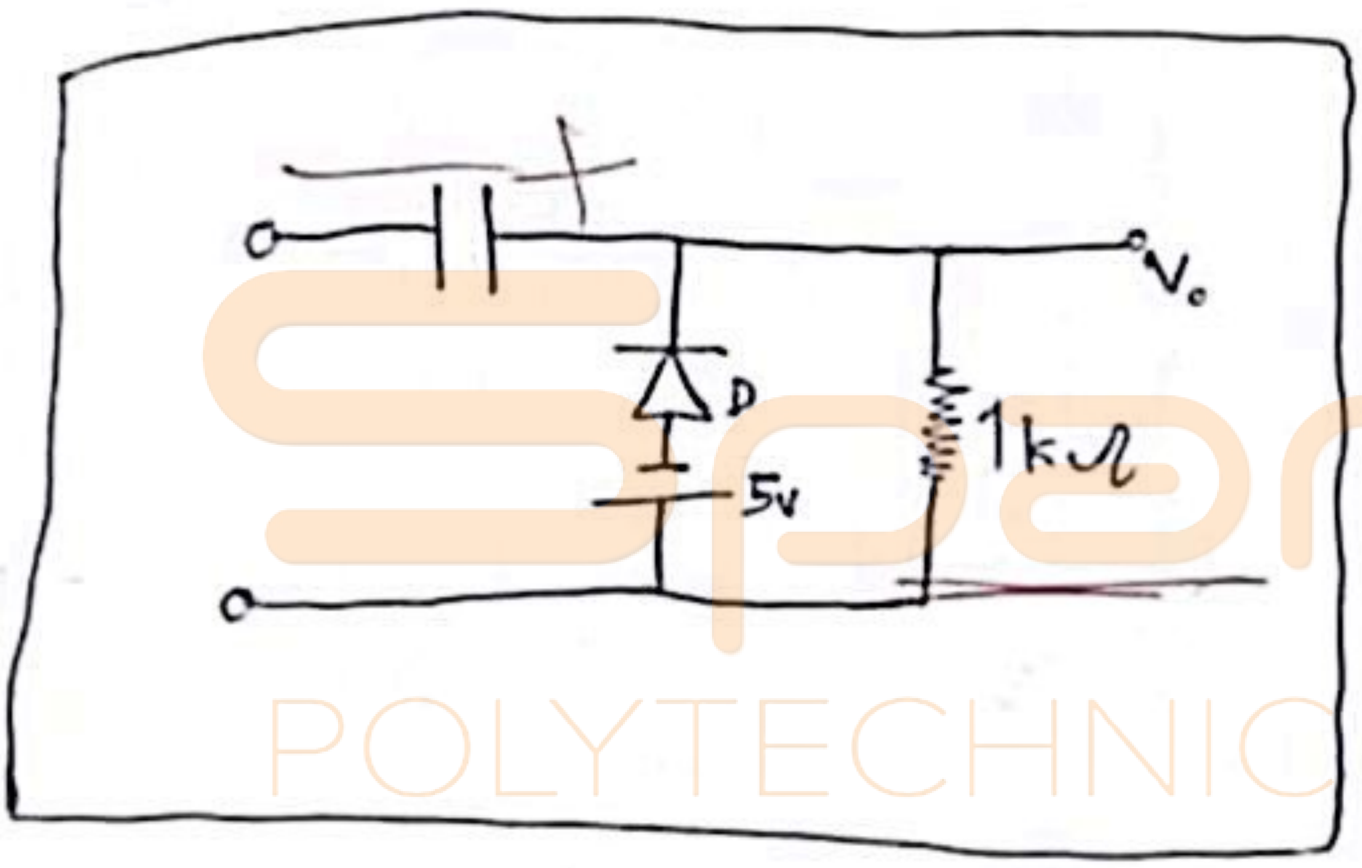
a- 4.54 mA b- 5.136 mA c-5.454 d- 0 mA

25- While making the test for the Zener diode circuit the output voltage received less than the Zener voltage in this case the Zener current is

a- $I_Z = I_{Zmax}$ b -$I_Z = I_R$ c- $I_Z = I_L$ $\boxed{I_Z = 0\ mA}$

Problem2 (5 points)

Design a clamper to perform the function indicated in figure shown (the **diode in the circuit is ideal)**



$\boxed{V_c = 15}$